

Josue Joel Villalobos Alferez

Backend Developer | Optimization & Applied AI

-  josue.villa778@gmail.com
-  4495803864
-  Mexico, Aguascalientes
-  [linkedin.com/in/josue-joel-villalobos-alferez-b5aa0a2b4](https://www.linkedin.com/in/josue-joel-villalobos-alferez-b5aa0a2b4)
-  github.com/josuejvillalobos

Summary

Software developer with 1.5 years of professional experience designing intelligent systems, microservice architectures, and data-driven pipelines. Skilled at translating complex operational problems into optimized, scalable software solutions. Multiple national hackathon winner with a strong track record of measurable impact across enterprise and field environments.

HARD SKILLS

- Development:** Java, JavaScript, TypeScript, Python, SQL, R, Spring Boot, React, Node.js, REST APIs, microservices
- AI & Data:** Machine learning, deep learning, Model Training, computer vision, metaheuristics, fuzzy logic, pandas/numpy, Power BI
- Infrastructure:** PostgreSQL, MongoDB, Redis, Docker, Git/GitHub
- Testing & Architecture:** JUnit 5, Mockito, Testcontainers, Postman, MVC, design patterns

Languages

- Spanish (Native)
- English (B2)
Oxford English Certification – In Progress
- German (Basic - A1)

Professional Experience

AI Developer – Intelligent Route Optimization System, MIAA

- 05/2026 – Present | Aguascalientes, Mexico
- Designed and deployed an AI-powered route optimization system for field notifiers, solving the TSP at operational scale, reducing average travel time by ~35% through metaheuristic algorithms applied to real-world geospatial data.
 - Optimized system performance through strategic caching of distance matrices, reducing redundant computation by ~70% under concurrent load.

Software Developer – HR & Organizational Systems, MIAA

- 04/2026 – 05/2026 | Aguascalientes, Mexico
- Architected an organizational management system covering the complete hierarchy of directorates, sub-directorates, departments, and positions, serving as a core internal tool for enterprise HR operations.
 - Reduced manual org-chart update time by ~60% through dynamic, role-based interfaces that enable real-time structural changes without backend intervention.

Automation Intern Nov 2024 – Mar 2025, Datallizer

- 11/2024 – 03/2025 | Aguascalientes
- Built an automated customs quotation management system, reducing manual processing time by ~30% and eliminating data entry errors across internal workflows.

Education

B.S. in Information Technology and Digital Innovation,

Metropolitan Technological University of Aguascalientes
09/2025 – Present | Aguascalientes, Mexico

Associate Degree (TSU) in Artificial Intelligence,

Metropolitan Technological University of Aguascalientes
08/2023 – 07/2025 | Aguascalientes, Mexico

Awards

1st Place – Environment Vertical "BioTerra", ICP. Medellin, Colombia

- 05/2024
- Developed a smart agriculture system with real-time soil monitoring and blockchain-based traceability.